LOCAL-FIRST DEVELOPER WORKSPACE / IMPLEMENTATION PLAN

Helpyard
One-Click Upgrade

Python bridge, automatic browser connection, and a single Windows double-click launcher

REVISION	DATE
2.0 - implementation complete	17 October 2026

DECISION SUMMARY

Keep the interface in HTML, CSS, and JavaScript. Run a dependency-free Python 3.10+ local bridge; retain fixed PowerShell subprocesses only for Windows-native operations. Use one .bat file to check Python, start the bridge minimized, and open the authenticated dashboard.

DELIVERY STATUS

IMPLEMENTED AND VERIFIED ON ONE WINDOWS WORKSTATION

No pip dependencies. No token saved in browser storage. No arbitrary shell or MCP process execution.

1. Executive summary

This revision records the selected architecture and the completed one-click migration. The Python bridge serves both the page and API on loopback, uses bounded request threads, and exchanges a single-use, short-lived bootstrap code for an in-memory browser token. A minimized console remains visible for startup/runtime diagnostics.

Outcome: The root launcher was run successfully. Ports 8787 and 8788 were occupied, so the bridge correctly selected 8789; its bridge probe returned HTTP 204 and the dashboard returned HTTP 200. A separate browser integration run verified automatic authentication and the live dashboard workflows.

Confirmed implementation decisions

Decision	Selected approach
Backend	Python standard library; minimum supported runtime is Python 3.10.
Frontend	Keep the existing single-file HTML/CSS/JavaScript dashboard; no framework/build migration.
Launcher	Root Start-Helpyard.bat; Python version check, minimized visible console, automatic browser launch.
Authentication	One-time code in the URL fragment, expires in 60 seconds, consumed once; bearer token stays in JS memory.
Operations	Windows features call fixed PowerShell scripts; /api/run and MCP startup remain disabled.
Dependencies	No pip packages are needed or installed; local Ollama remains optional.

Scope boundaries

The bridge binds to IPv4 loopback only. The project remains a local prototype, not a hosted/multi-user product. Cloud AI still requires the user's own provider key; it is entered in the browser and not persisted. Destructive actions were not executed during verification.

2. Target architecture and data flow

The runtime has one browser origin and one local Python process. Python validates host/origin, authentication, request shape, and explicit approvals. Windows data/actions are delegated to fixed, encoded PowerShell scripts; Ollama calls are restricted to localhost.

Stage	Behavior
1 - Launch	Double-click Start-Helpyard.bat; verify Python >= 3.10.
2 - Serve	Bind 127.0.0.1; select first available configured port; start bounded HTTP workers.
3 - Connect	Open browser with short-lived code in URL fragment; fragment is removed before POST exchange.
4 - Authorize	Return random bearer token once; keep only in page memory; require it for protected APIs.
5 - Operate	Load local data; make fixed Windows/Ollama calls; show errors; require API approval for mutations.

Reliability and security properties

Control	Implementation
Loopback boundary	Listens on 127.0.0.1; Host, client address, and browser Origin are validated.
Bootstrap lifecycle	60-second expiry, same-origin restriction, constant-time comparison, one successful redemption.
Credential handling	No bearer token in query, localStorage, or normal console logs; fragment removed before code exchange.
Request controls	1 MiB JSON body limit, schema/field validation, 16-worker cap, overload response is HTTP 503.
Operations	Mutation routes require explicit server approval; arbitrary PowerShell and MCP startup return HTTP 403.
Slow AI	Independent request workers keep ping responsive while Ollama chat is waiting.

Trust warning: A local bridge token permits access to workstation data and reviewed named operations. Do not expose the port to a LAN/public network or share the temporary connection URL.

3. Step-by-step implementation plan

The requested implementation was delivered in dependency order. Completion means code and measurable checks exist; it does not imply public-release certification or mutation testing.

Step	Work	Status	Acceptance / evidence
1. Baseline and failure analysis	Preserve dashboard UX; identify offline/401 conditions and serial-bridge timeout during long Ollama inference.	**DONE**	Existing UI retained; Python concurrency regression test proves ping responds during simulated slow chat.
2. Python bridge	Serve dashboard/API; preserve route shapes; bind loopback; validate inputs, origins, approvals, and local Ollama URLs.	**DONE**	py_compile passes; authenticated live metrics, projects, process/service inventories, and Ollama requests work on this host.
3. Automatic connection	Mint one-time bootstrap code and memory-only bearer token; remove fragment before exchange; reject expiry/replay/wrong origin.	**DONE**	HTTP tests cover expiry, replay, invalid code, and origin; browser URL cleans itself and stays Connected.
4. One-click launcher	Check Python version; start bridge in minimized visible console; open actual selected-port URL; handle port fallback.	**DONE**	Start-Helpyard.bat smoke run used fallback port 8789; dashboard HTTP 200 and probe HTTP 204.
5. Test and UX wiring	Check routes, API schemas, approvals, all 12 navigation sections, filters, no token persistence, slow-AI availability.	**DONE**	Six Python tests; Node check: 12 nav targets, 10 API routes, 78 DOM IDs; browser listed 60 processes, 296 services, 15 Ollama models; chat returned READY.
6. Docs and release gate	Document startup, security limits, verification boundaries, and next-phase tasks; update the plan PDF.	**DONE**	README, VS Code task/launch settings, project report, and this revised PDF describe the selected Python path.

Operator acceptance checklist

	Acceptance item
[x]	Double-click launcher starts without installing pip packages.
[x]	Bridge serves page and API on a selected loopback port and opens browser.
[x]	Browser connects without copying a token; token is not persisted.
[x]	All 12 sections render and bridge-backed operational panels populate.
[x]	Slow local chat does not make bridge health/ping go Offline.
[ ]	Repeat on Python 3.10 and supported installed browsers before wider release.
[ ]	Do not test stop/restart/shutdown/service mutations outside an approved test plan.

4. Verification results

Checks were run against the current Python implementation on Windows 11 with Python 3.14, Node.js 24, and the integrated Chromium browser. Legacy PowerShell tests are distinguished from the default Python path.

Area	Result	Evidence
Python syntax	**PASS**	py -3 -m py_compile bridge.py tests\test_bridge.py
Python API suite	**PASS - 6 tests**	Auth, bootstrap origin/expiry/replay, route schemas, approval gates, malformed/oversized bodies, slow-chat concurrency.
Dashboard static checks	**PASS**	12 navigation targets, 10 API routes, 78 IDs, JavaScript syntax; no missing local assets.
One-click batch	**PASS**	Exited 0; port 8789 selected after 8787/8788 were busy; served root page 200 and probe 204.
Browser bootstrap	**PASS**	Connected automatically from #connect fragment; visible URL no longer contained code; persisted localStorage did not contain token.
Navigation and live data	**PASS**	All 12 pages visible; overview metrics and disks; project list; 60 process rows; 296 service rows; filter reduced process list.
Ollama	**PASS**	15 models listed; local response READY; sidebar stayed Connected.
Unsafe actions	**PASS**	/api/run and MCP start rejected; mutation API approval cases return 403 in tests. No destructive action was executed.

Known verification limits

• Only one Windows 11 workstation and Python 3.14 were exercised; Python 3.10 minimum compatibility has not yet been run.

• Browser automation covered one integrated Chromium-based browser; Edge/Chrome/Firefox release matrix and accessibility checks remain outstanding.

• PowerShell-backed list/read routes were exercised live. Service/process/power mutations, Claude config writes, and admin-only actions were not run.

• Ollama used the installed local models; no throughput, quality, or high-load test was performed.

• No security certification, external network exposure test, public deployment, or adversarial penetration review was performed.

5. Start, recover, and continue safely

Normal startup

Item	Instruction
Prerequisite	Python 3.10+ installed; Windows PowerShell 5.1 available for native operations; Ollama optional.
Start	Double-click Start-Helpyard.bat at the repository root. VS Code task: Terminal > Run Task > Helpyard: Start Bridge.
Keep running	Leave minimized Helpyard console open. It is the diagnostic surface; close it or press Ctrl+C to stop.
No packages	The bridge uses Python standard library only; there is no pip install step.
Manual fallback	Run py -3 .\bridge.py --open-browser from this project directory.

Recovery map

Symptom	Action
Python missing/old	Install Python 3.10+ for Windows; ensure py -3 or python works from a new terminal.
Port already used	No manual port change is normally needed; server tries its configured fallback list and prints the selected port.
401 / expired startup code	Close bridge and run launcher again; each code is single-use and valid for 60 seconds.
Bridge stopped	Keep the minimized console open; restart launcher to create a fresh process/session.
Ollama unavailable	Dashboard remains usable; start Ollama at 127.0.0.1:11434 and refresh models.
PowerShell route error	Read the bridge console diagnostics; verify Windows PowerShell 5.1 is present and the operation is allowed.

Next work before broader release

Priority	Follow-up
P1	Run Python 3.10 compatibility checks; exercise supported installed browsers and smaller viewports.
P1	Add automated launcher checks and Windows-specific integration tests that do not mutate user state.
P1	Review output caps, subprocess cancellation and error surfaces; complete threat-model and dependency/runtime review.
P2	Add persistent audit events and explicit workspace approval; keep arbitrary terminal and MCP startup disabled unless separately reviewed.
P3	Use a mock-only backend for any public demo; never connect public clients to the workstation bridge.

Release gate: Local usage is verified for this machine. Wider distribution remains blocked on supported-runtime/browser coverage, a complete security review, and documented operator support. This PDF records evidence and limitations; it is not a guarantee of zero defects.